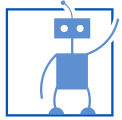


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Reinforcement Learning: Overview & Comparison, Approximative RL

Prof. Dr.-Ing. habil. Alois Knoll

Mohammadhossein Malmir, M.Sc.

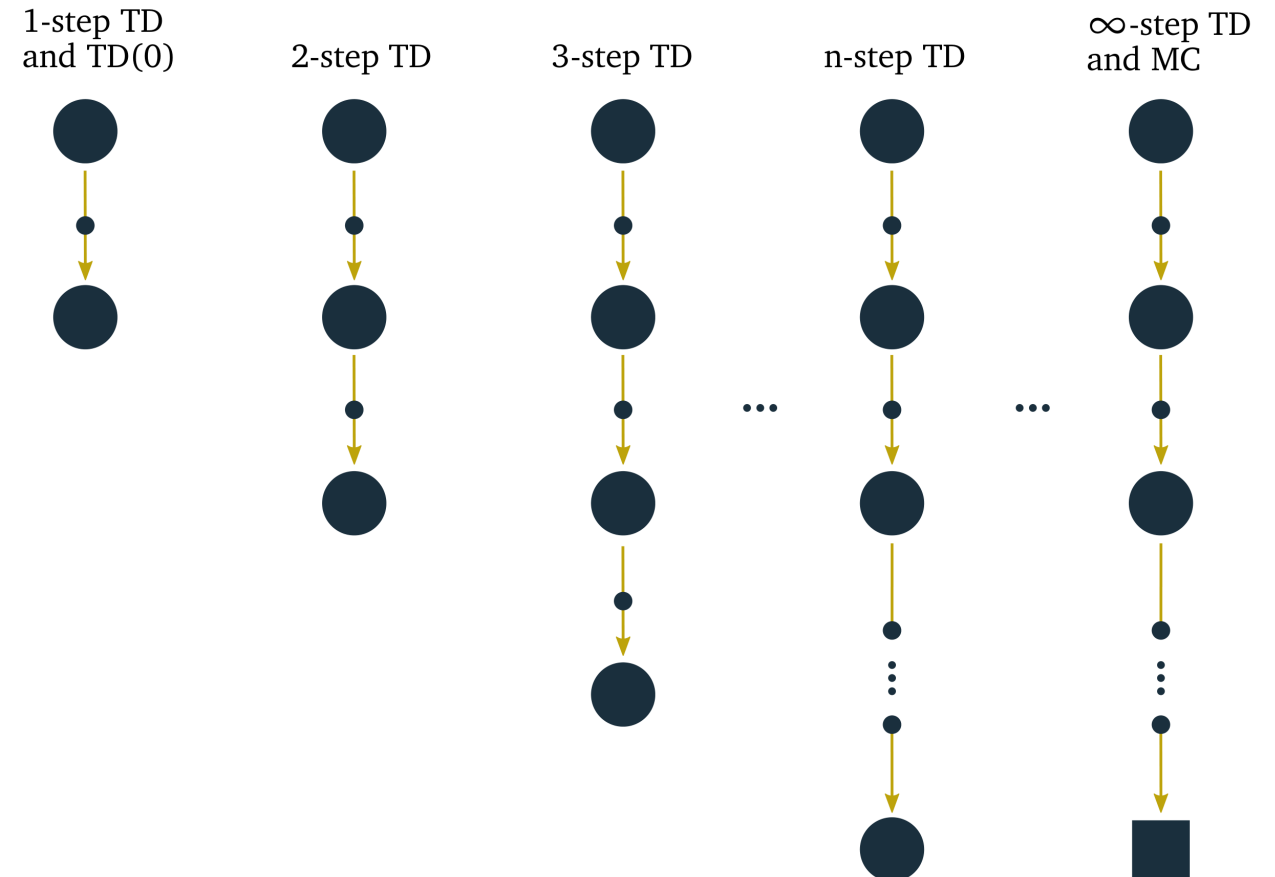
Kejia Chen, M.Sc.

Agenda

- **Overview & Comparison**
 - N-step Bootstrapping
 - Comparison of DP, MC, and TD
- Approximate RL

n-Step Bootstrapping[§]

- The n -step TD algorithm incorporates n samples before bootstrapping value.
- This approach unifies TD and MC:
 - $n = 1 \rightarrow \text{TD}$
 - $n = \infty \rightarrow \text{MC}$



[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, **p.142**.
Slide is adapted from CogSys21 RL lecture held by Prof. Dr. Ing. Noah Klarmann

n-Step Bootstrapping[§]

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$$V(S_t) \leftarrow V(S_t) + \alpha(G_t - V(S_t))$$

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- one-step TD $G_{t:t+1} \doteq R_{t+1} + \gamma V_t(S_{t+1})$
- two-step TD $G_{t:t+2} \doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 V_{t+1}(S_{t+2})$
- n-step TD $G_{t:t+n} \doteq R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{n-1} R_{t+n} + \gamma^n V_{t+n-1}(S_{t+n})$
- MC $G_{t:t+n} \doteq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} \dots + \gamma^{T-t-1} R_T$

Overview & Comparison

- **Key ideas in common***
 1. Estimate value functions
 2. Back up value updates
 3. Generalized policy iteration

- **Sample updates? §**
update samples an expectation

DP	MC	TD
✗	✓	✓

- **Bootstrapping? §**
update involves an estimate

DP	MC	TD
✓	✗	✓

§David Silver. Reinforcement Learning: UCL Course, 2015, 4th lecture, slide 29.

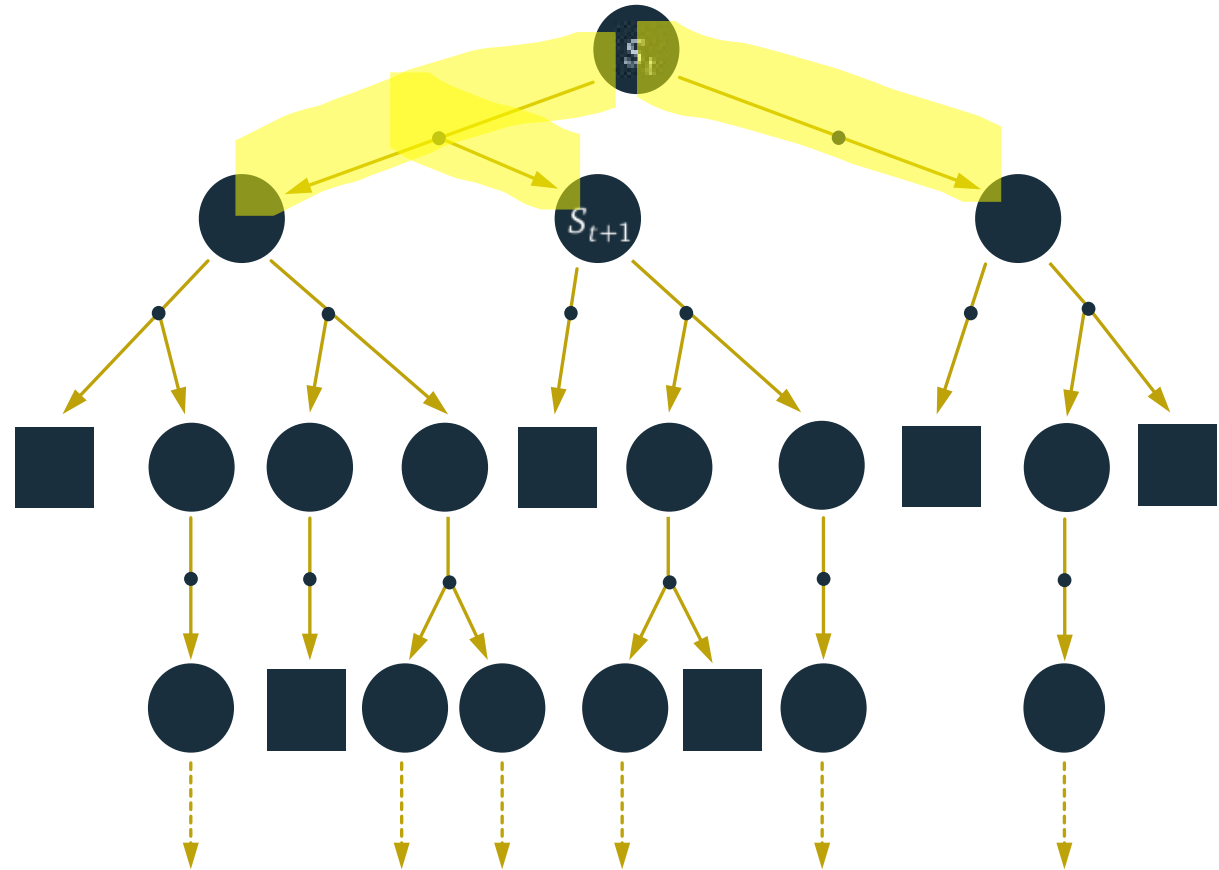
*Sutton, R. S. and Barto, A. G. Reinforcement learning: An introduction. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.190.

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Overview & Comparison

- Dynamic Programming Backup

$$V(S_t) \leftarrow \mathbb{E}_{\pi}[R_{t+1} + \gamma V(S_{t+1})]$$

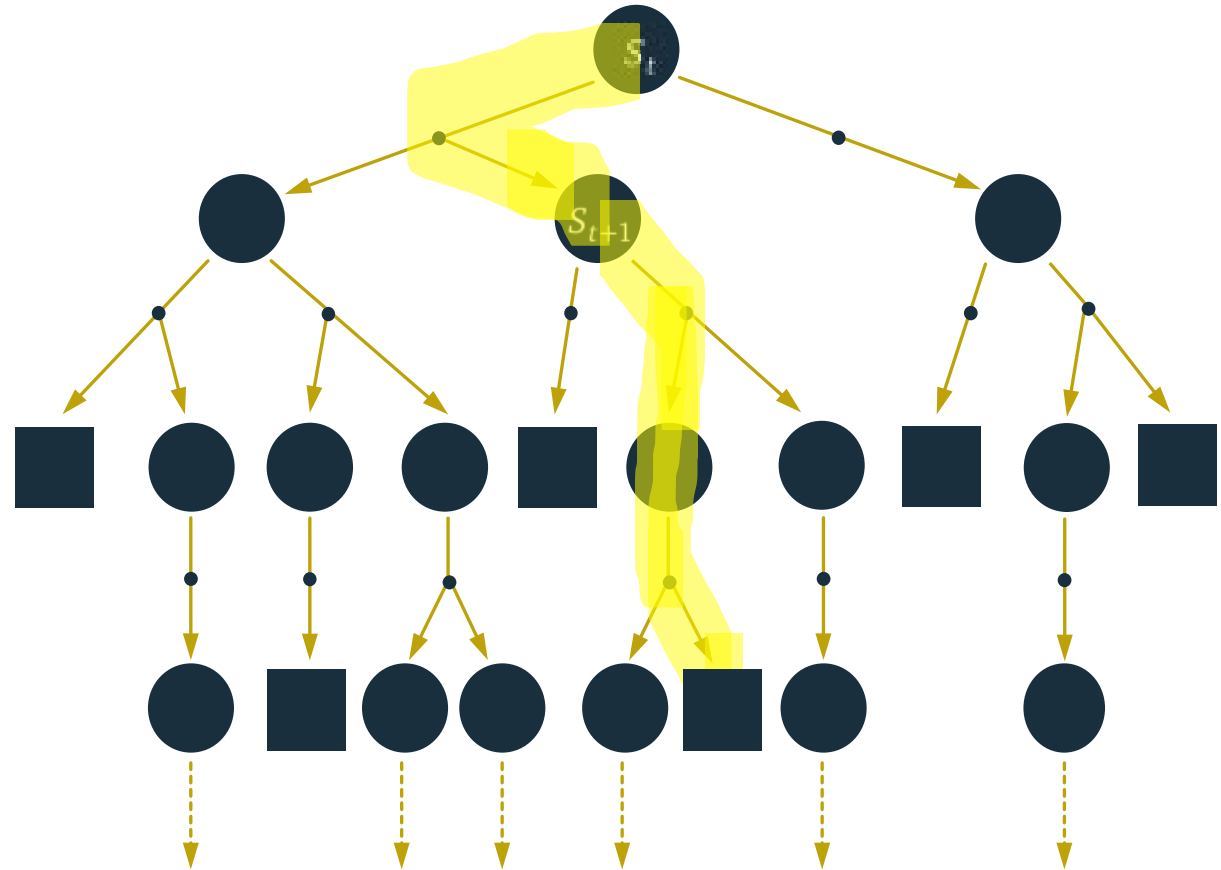


^sDavid Silver. Reinforcement Learning: UCL Course, 2015, 4th lecture, slide 28.

Overview & Comparison

- Monte Carlo Backup

$$V(S_t) \leftarrow V(S_t) + \alpha(G_t - V(S_t))$$

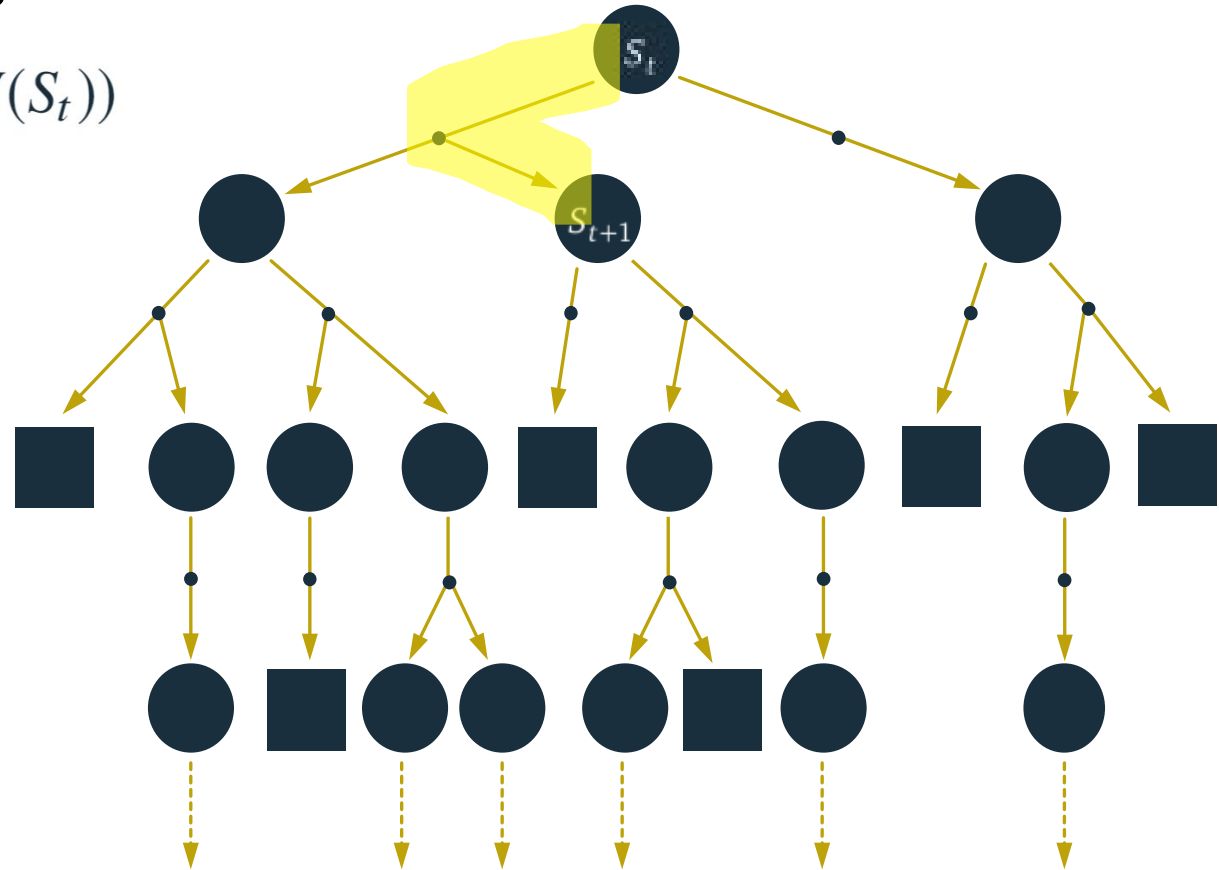


[§]David Silver. Reinforcement Learning: UCL Course, 2015, 4th lecture, slide 26.

Overview & Comparison

- One-step Temporal Difference Backup

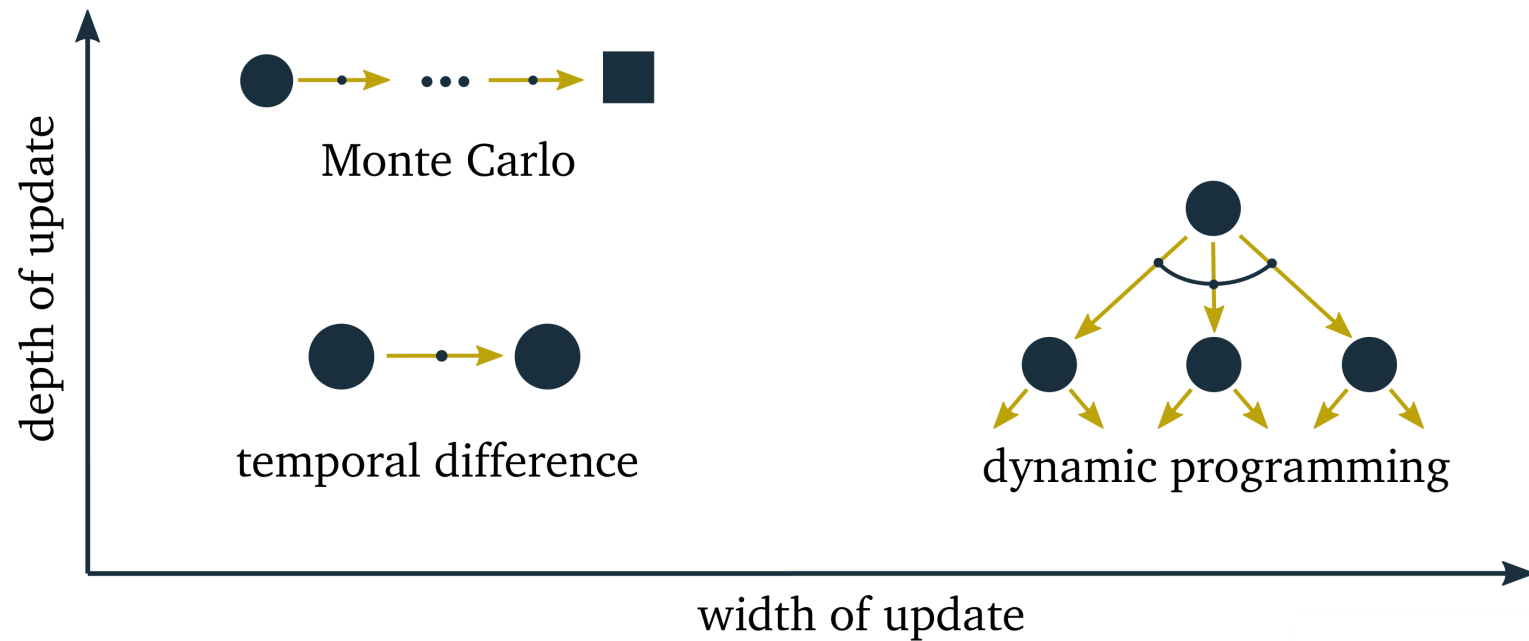
$$V(S_t) \leftarrow V(S_t) + \alpha(R_{t+1} + \gamma V(S_{t+1}) - V(S_t))$$



[§]David Silver. Reinforcement Learning: UCL Course, 2015, 4th lecture, slide 27.

Overview & Comparison

- Depth and width of update:*



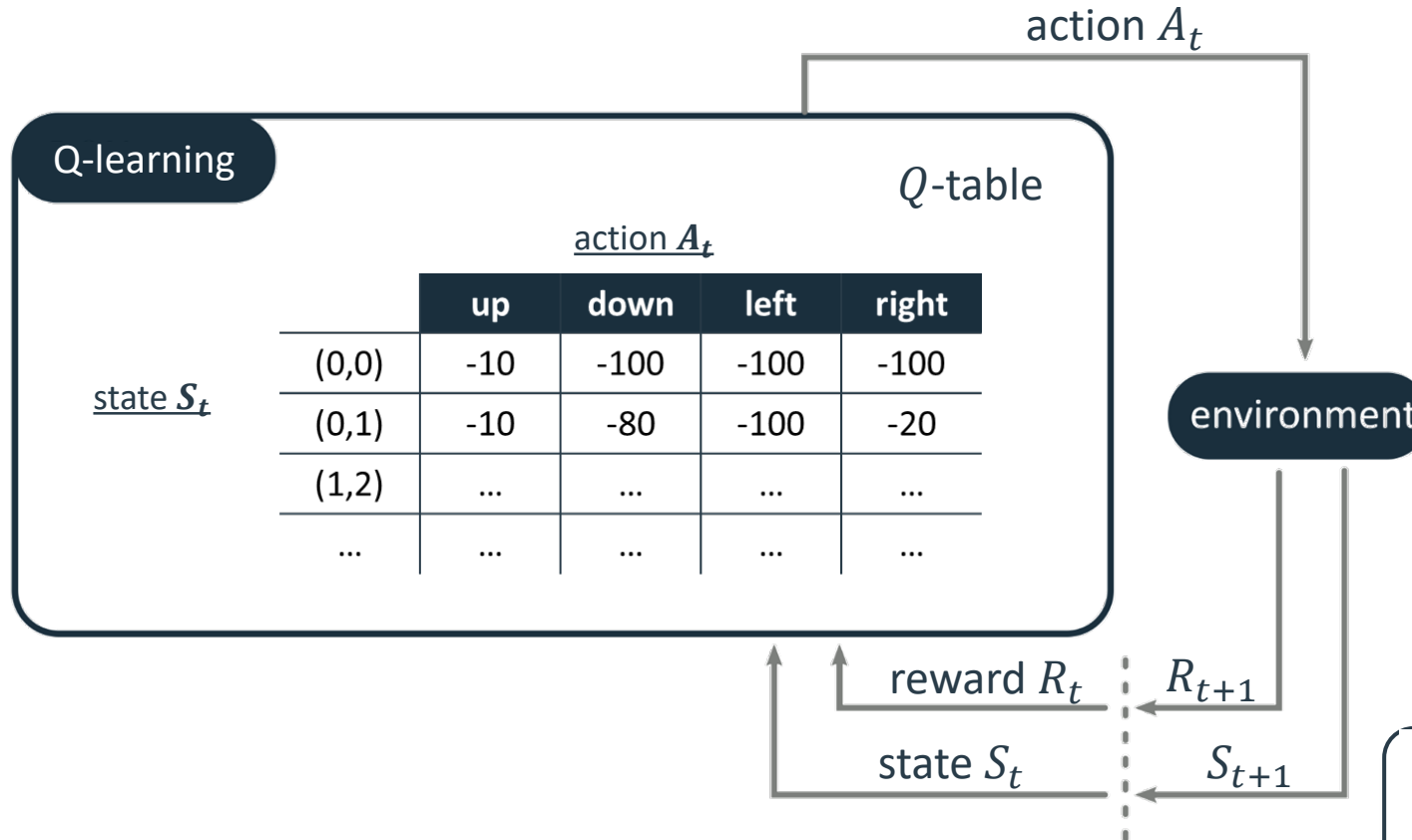
**Sutton, R. S. and Barto, A. G. Reinforcement learning: An introduction. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.190.*

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Agenda

- ✓ Overview & Comparison
 - ✓ N-step Bootstrapping
 - ✓ Comparison of DP, MC, and TD
- **Approximate RL**

Q-table

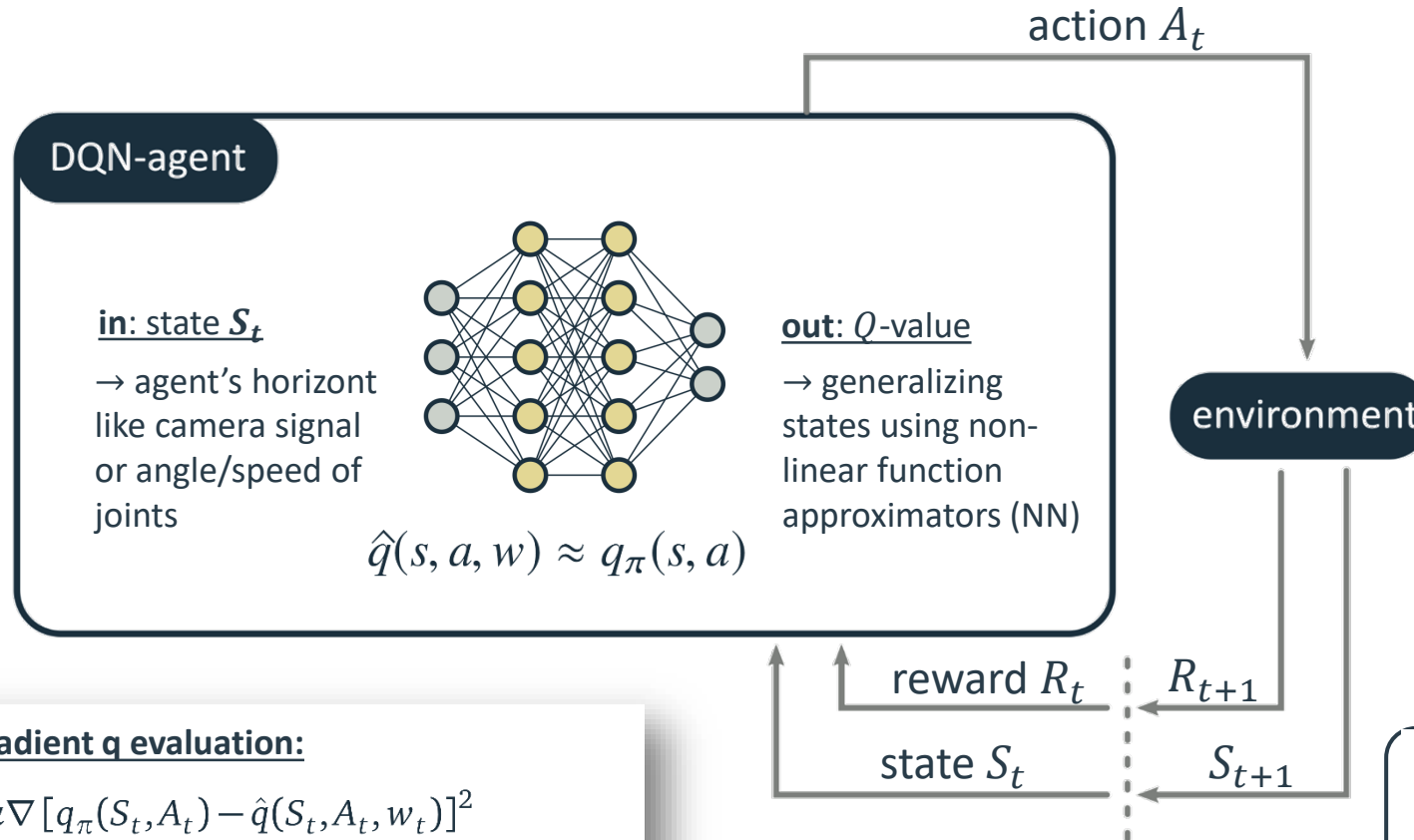


[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.201.

Slide is adapted from CogSys21 RL lecture held by Prof. Dr. Ing. Noah Klarmann

α	learning rate
γ	discount rate
A	stochastic action function
q	value for a specific action
R	stochastic reward function
S	stochastic state function

Approximative RL: Deep Q-Network[§]



[§]Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.201.

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SGD for semi-gradient q evaluation:

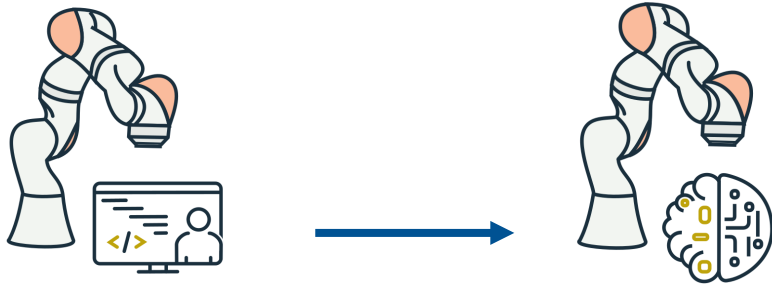
$$w_{t+1} = w_t - \frac{1}{2} \alpha \nabla [q_\pi(S_t, A_t) - \hat{q}(S_t, A_t, w_t)]^2$$

$$= w_t + \alpha [q_\pi(S_t, A_t) - \hat{q}(S_t, A_t, w_t)] \nabla \hat{q}(S_t, A_t, w_t)$$

α	learning rate
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A	stochastic action function
q	value for a specific action
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RL for Robotic Manipulation: AI4DI Vision

Vision:



Current State:

Applying hand-crafted rules to control industrial robots.

Goal State:

Employing self-taught action policies to control industrial robots.

Implication:



increased **performance**



adaption to product variations



reduced **product batch sizes**



increased **safety**



decreased **costs**

Slide is adapted from CogSys21 RL lecture held by Prof. Dr.-Ing. Noah Klarmann.

This Makes Reinforcement Learning Difficult...[§]

Robot Grasping Experiment by Google

Learning to grasp by large-scale data collection with **physical** robots:

- Up to **14 robots** operating in parallel
- **Two months** runtime
- **800,000** grasp attempts

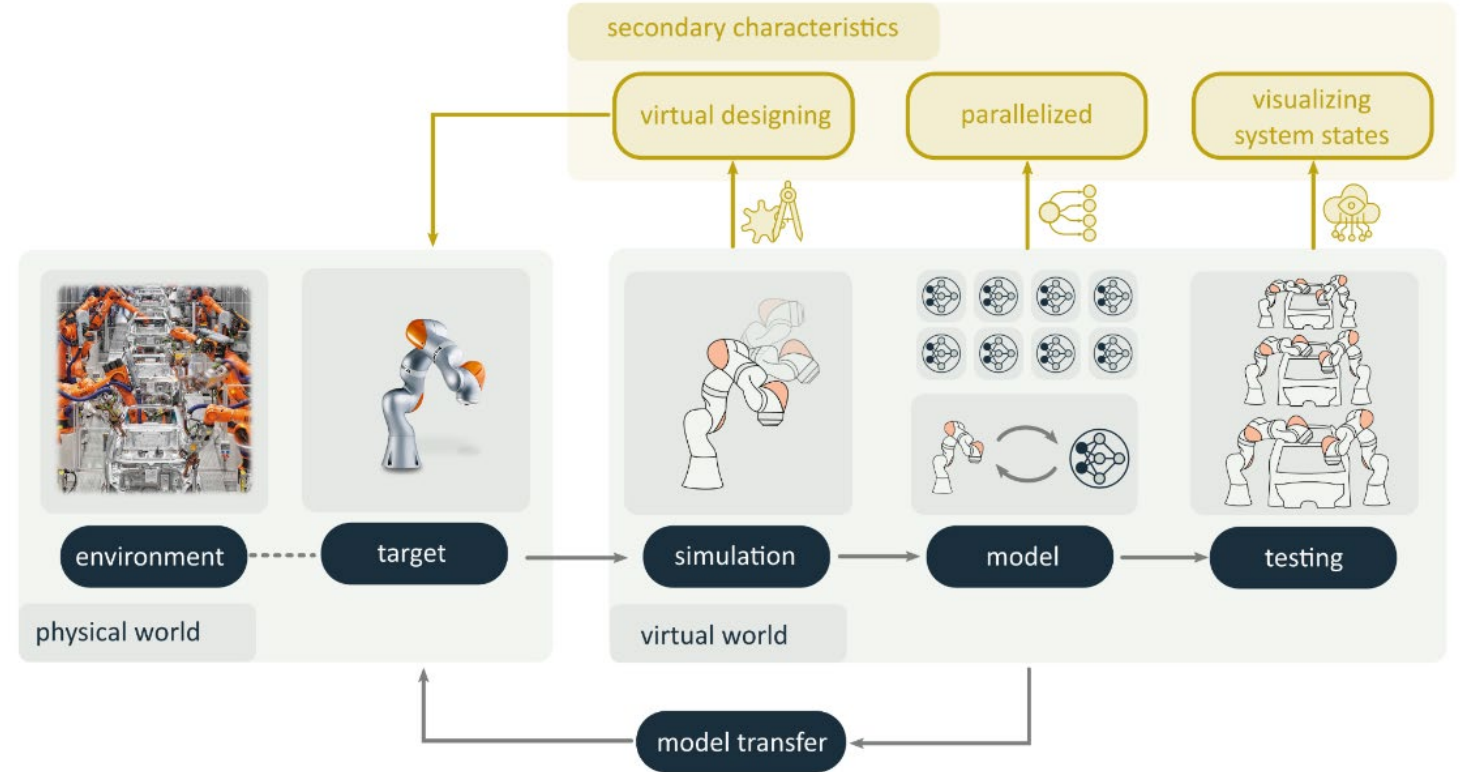
[§]S. Levine, P. Pastor, A. Krizhevsky, J. Ibarz, and D. Quillen. "Learning hand-eye coordination for robotic grasping with deep learning and large-scale data collection". In: *The International Journal of Robotics Research* 37.4-5 (2018), pp. 421–436.



RL for Robotic Manipulation: AI4DI Mission^{§,*}

Solution:

Training digital twins in virtual environments and transfer them to control the real systems.



Slide is adapted from CogSys21 RL lecture held by Prof. Dr.-Ing. Noah Klarmann.

[§]J. Josifovski, M. Malmir, N. Klarmann, A. Knoll. "Continual Learning on Incremental Simulations for Real-World Robotic Manipulation Tasks". In: 2nd Workshop on Closing the Reality Gap in Sim2Real Transfer for Robotics (2020).

^{*}M. Malmir, J. Josifovski, N. Klarmann, A. Knoll. "Robust Sim2Real Transfer by Learning Inverse Dynamics of Simulated Systems". In: 2nd Workshop on Closing the Reality Gap in Sim2Real Transfer for Robotics (2020).

RL for Robotic Reaching Task[§]

example task: reaching

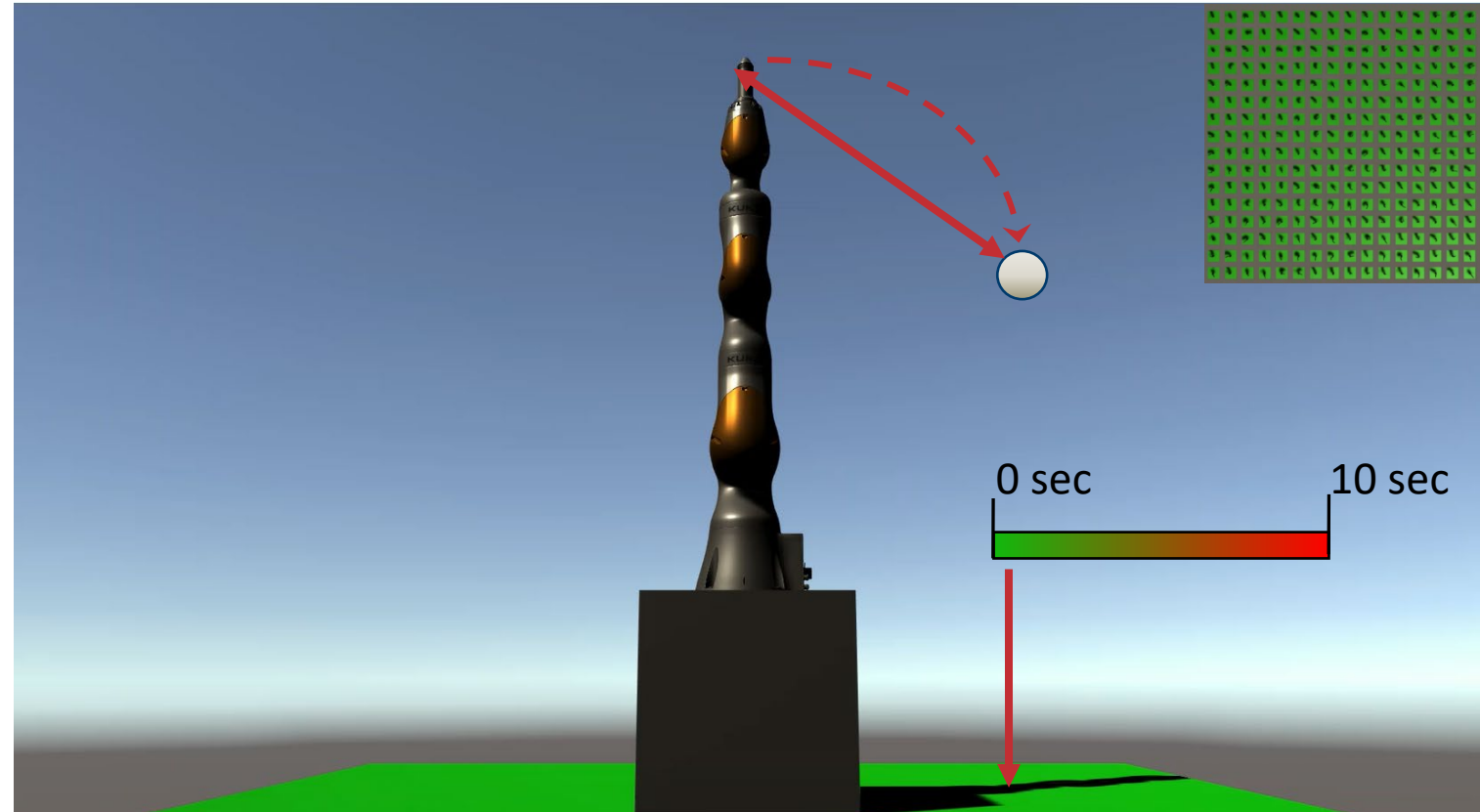
frequency: 50 Hz

observation: distance from target position ($\Delta x, \Delta y, \Delta z$) and current angle and speed of the first 4 joints in deg/s

action: velocity reference for each of the 4 joints

reward: - distance; +10 for reaching

time: 10 seconds (500 steps)



[§]J. Josifovski, M. Malmir, N. Klarmann, A. Knoll. "Continual Learning on Incremental Simulations for Real-World Robotic Manipulation Tasks". In: 2nd Workshop on Closing the Reality Gap in Sim2Real Transfer for Robotics (2020).

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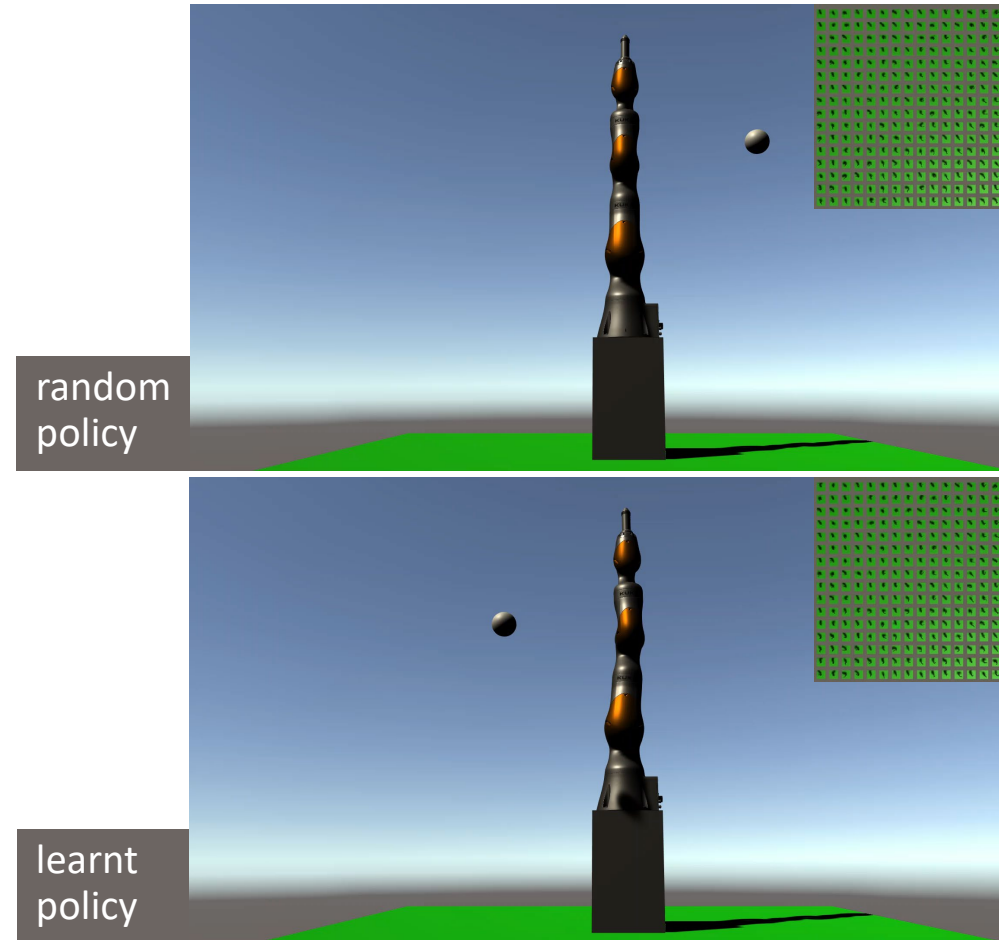
frequency: 50 Hz

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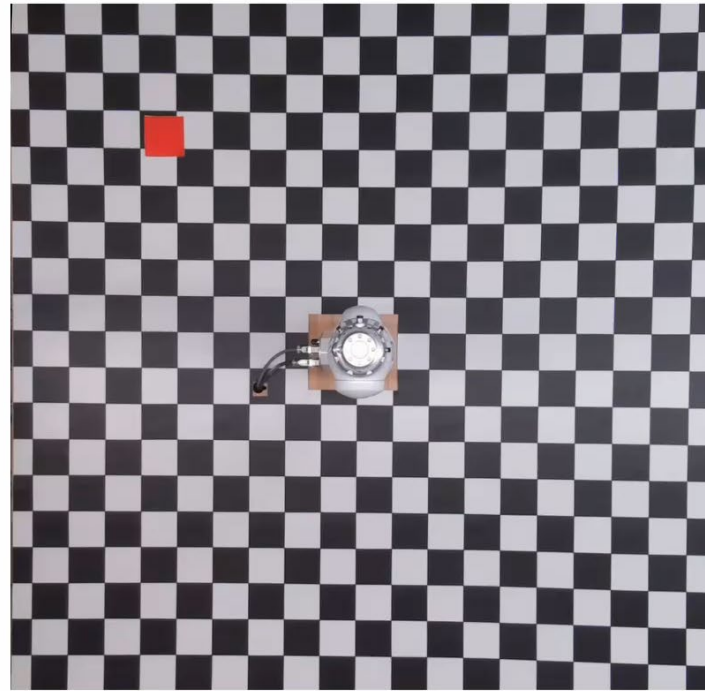
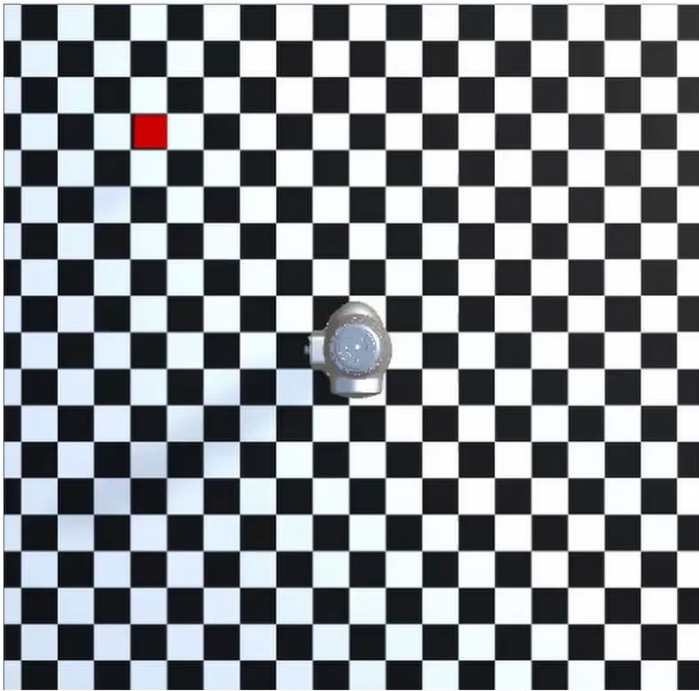
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[§]J. Josifovski, M. Malmir, N. Klarmann, A. Knoll. "Continual Learning on Incremental Simulations for Real-World Robotic Manipulation Tasks". In: 2nd Workshop on Closing the Reality Gap in Sim2Real Transfer for Robotics (2020).

RL for Robotic Reaching Task Sim2Real Transfer[§]

Evaluation of simulation-only trained model in simulated (left) and corresponding real environment (right). The simulation-trained model is successfully transferred to control a robot in the real world.



Reaching task, max joint speed: 5 deg/sec, video speed: 4.0 x real-time

[§]Josifovski, J., Malmir, M., Klarmann, N., Žagar, B. L., Navarro-Guerrero, N., & Knoll, A.: "Analysis of Randomization Effects on Sim2Real Transfer in Reinforcement Learning for Robotic Manipulation Tasks". In 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (2022).

Thank you!